Lesson 6.7 **Duration: 40 minutes**

Cloud Services

Learning Objectives

- Compare major cloud platforms and their service offerings
- Identify appropriate cloud services for different application needs
- Evaluate cloud service reliability, security, and cost considerations

Engage (5-7 minutes)

Show logos of major cloud providers: Amazon Web Services, Microsoft Azure, Google Cloud Platform. Ask students which services they already use from each (AWS hosts Netflix, Azure powers Microsoft 365, Google Cloud runs YouTube). Discuss how these companies have become the infrastructure backbone of the internet.

Explore (10-12 minutes)

Students receive a project brief: build a mobile app for a local business. They must select cloud services for: hosting the backend (compute), storing user data (database), sending notifications (messaging), handling payments (payment processing), and storing images (object storage). They compare offerings from two providers and make justified selections.

Explain (10-12 minutes)

Major cloud providers offer comprehensive ecosystems. AWS provides 200+ services including EC2 (compute), S3 (storage), RDS (databases), and Lambda (serverless). Azure offers Virtual Machines, Blob Storage, SQL Database, and integrates deeply with Microsoft tools. Google Cloud provides Compute Engine, Cloud Storage, BigQuery (data analytics), and TensorFlow (machine learning). Each provider has strengths: AWS for breadth, Azure for enterprise integration, Google Cloud for data analytics and AI. Serverless computing (AWS Lambda, Azure Functions) lets developers run code without managing servers.

Elaborate (8-10 minutes)

Analyze cloud service reliability: Service Level Agreements (SLAs) guarantee 99.9% to 99.999% uptime. Discuss multi-cloud strategies—using multiple providers to avoid vendor lock-in and increase resilience. Examine security considerations: shared responsibility model, encryption, identity management, and compliance certifications. Cost optimization: rightsizing instances, using spot instances for fault-tolerant workloads, and leveraging reserved pricing for predictable loads.

Evaluate (5-8 minutes)

Students create a comparison matrix of cloud services for a specific use case, evaluating price, performance, features, and support. They must select a provider and justify their choice. Case study analysis: examine a real cloud outage (e.g., AWS us-east-1 incidents) and discuss its impact and lessons learned.